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| United States Patent | 12404218                |
| Kind Code            | B2                      |
| Date of Patent       | September 02, 2025      |
| Inventor(s)          | Blair; Taylor K. et al. |

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### Article with surface structures for CMAS resistance

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#### Abstract

In some examples, article used as a component for a turbine engine that operates in a high temperature environment. The article may include: a ceramic or ceramic matrix composite (CMC) substrate; and a coating on the ceramic or the CMC substrate, wherein the coating defines an outer surface of the article. The coating includes a plurality of surface features defining channels on the outer surface of the article. The channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article.

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**Inventors:** Blair; Taylor K. (Indianapolis, IN), Freeman; Ted J. (Indianapolis, IN), Sippel; Aaron (Indianapolis, IN)

**Applicant:** Rolls-Royce Corporation (Indianapolis, IN)

**Family ID:** 86992361

**Assignee:** Rolls-Royce Corporation (Indianapolis, IN)

**Appl. No.:** 17/646551

**Filed:** December 30, 2021

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20230212086 A1          | Jul. 06, 2023           |

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#### Publication Classification

**Int. Cl.:** C04B41/50 (20060101); C04B35/80 (20060101); C04B41/45 (20060101); C04B41/87 (20060101)

## U.S. Cl.:

CPC     **C04B41/5024** (20130101); **C04B35/80** (20130101); **C04B41/459** (20130101);  
**C04B41/87** (20130101); C04B2235/9669 (20130101)

## Field of Classification Search

**CPC:**     C04B (41/5024); C04B (35/80); C04B (41/459); C04B (41/87); C04B (2235/9669);  
C04B (41/009); C04B (41/5042); C04B (35/573); C04B (41/53); C04B (2103/0021);  
F23R (3/007); F05D (2240/11); F05D (2240/12); F05D (2240/30); F05D (2250/11);  
F05D (2250/181); F05D (2250/182); F05D (2250/183); F05D (2250/21); F05D  
(2250/23); F05D (2300/6033); F01D (5/282); F01D (5/284); F01D (5/288); F01D  
(25/007); Y02T (50/60)

**USPC:**   428/156; 428/167; 428/172; 428/192; 428/212; 428/213

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*Primary Examiner:* Gaitonde; Megha M

*Attorney, Agent or Firm:* Shumaker & Sieffert, P.A.

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## Background/Summary

## TECHNICAL FIELD

(1) The present disclosure relates to high temperature components such as components of gas turbine engine that operate in high temperature environments.

## BACKGROUND

(2) The components of gas turbine engines operate in severe environments. For example, some components exposed to hot gases in commercial aeronautical engines may experience surface temperatures in excess of about 1200° C. At these extreme temperatures, component substrates may be exposed to environmental species such molten Calcium-Magnesium-Alumino-Silicate (CMAS) containing materials, which can cause chemical and/or mechanical damage to parts. Improving component resistance to molten CMAS has involved a focus on coating chemistry such as, for example, development of environmental barrier coatings (EBCs). Engine components may be coated with one or more barrier layers to provide protection against thermal flux, erosion, and/or environmental contamination, for example, by reducing or preventing CMAS formation, migration, or infiltration.

## SUMMARY

(3) In some examples, the disclosure is directed to an article comprising: a ceramic or ceramic matrix composite (CMC) substrate; and a coating on the ceramic or the CMC substrate, wherein the coating defines an outer surface of the article, wherein the coating includes a plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article.

(4) In some examples, the disclosure is directed to a method comprising forming a coating on a ceramic or ceramic matrix composite (CMC) substrate, wherein the coating defines an outer surface of the article, wherein the coating includes a plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article.

(5) The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a conceptual schematic diagram illustrating a top view of an example article including a plurality of surface features in accordance with some examples of the disclosure.

(2) FIG. 2A is a conceptual schematic diagram illustrating an example side view of the example article of FIG. 1, in accordance with some examples of the present disclosure.

(3) FIG. 2B is a conceptual schematic diagram illustrating alternative example side view of the example article of FIG. 1, in accordance with some examples of the present disclosure.

(4) FIG. 2C is a conceptual schematic diagram illustrating alternative example side view of the example article of FIG. 1, in accordance with some examples of the present disclosure.

(5) FIG. 3 is a conceptual schematic diagram illustrating a top view of another example article including a plurality of surface features in accordance with some examples of the disclosure.

(6) FIG. 4 is a conceptual schematic diagram illustrating an example side view of the example article of FIG. 3, in accordance with some examples of the present disclosure.

(7) FIG. 5 is a conceptual schematic diagram illustrating a top view of another example article

including a plurality of surface features in accordance with some examples of the disclosure.

(8) FIG. 6 is a conceptual schematic diagram illustrating an example side view of the example article of FIG. 5, in accordance with some examples of the present disclosure.

(9) FIG. 7 is a conceptual schematic diagram illustrating a top view of another example article including a plurality of surface features in accordance with some examples of the disclosure.

(10) FIG. 8 is a flowchart illustrating an example technique for forming an article including a coating on a substrate, where the coating includes surface features in accordance with some examples of the present disclosure.

(11) FIG. 9 illustrates an example CMAS resistant coating after exposure molten CMAS for eight hours, showing CMAS pooled on the surface and not penetrating the coating.

(12) FIG. 10 illustrates the CMAS resistant coating of FIG. 9 after 96 hours, showing the molten CMAS beginning to react with and penetrate the coating.

(13) FIG. 11 illustrates an example CMAS susceptible coating with the molten CMAS absorbed into the coating.

(14) FIG. 12 is a closeup illustrating the CMAS susceptible coating of FIG. 9, showing the created microstructure and penetration of molten CMAS.

(15) Like symbols in the drawings indicate like elements.

#### DETAILED DESCRIPTION

(16) In some examples, the disclosure relates to articles that may be employed as components in high temperature mechanical systems, such as gas turbine engines, and techniques for making and using the same. In gas turbine engines, increasing demands for greater operating efficiency (e.g., fuel efficiency) has led to the operation of gas turbine engines at higher temperatures. Ceramic matrix composite (CMC) components are making their way into gas turbine engines due to their potential for higher operating temperature relative to other materials. However, CMC components in gas turbine engines are also susceptible to damage from CMAS, and the higher operating temperatures at which CMCs operate the potential damage from CMAS can shorten component life or cause catastrophic failure of a part. Since CMAS melts at the higher operating temperatures of CMC components, the molten CMAS can react more readily with protective environmental coating layers, which has constrained operating temperatures or required layered coatings to resist or delay the chemical or mechanical damage CMAS causes coatings and parts. Given sufficient time in contact with a coating in a specific region of a turbine engine, the molten CMAS can infiltrate and/or react with the CMC coating. However, as described below with regard to FIGS. 9-12, for example, there may be time to remove molten CMAS from the surface of a coated CMC before the CMAS completely reacts with a CMC component.

(17) Examples of the present disclosure include articles that incorporate structure onto gas washed outer surfaces to promote the removal of CMAS from their surface. In some examples, the articles may include a ceramic or CMC substrate having a coating such as an environmental barrier coating and/or abradable coating defining the outer surface of the component. The surface of the coating may include one or more protruding surface features and/or channels that promote the flow of a molten material such as molten CMAS within a gas stream downstream, e.g., so that the molten CMAS is transferred downstream of the component rather than pooling or otherwise gathering on the surface of the component. Such structures may decrease CMAS exposure in the hottest portions of a gas turbine engine, pushing the CMAS downstream (and possibly beneficially atomizing the fluid) to lower temperature areas where the CMAS solidifies or may be more easily managed as lower temperatures slow reaction rates. The structures may reduce accumulation of molten materials on critical components and reduce stresses during freeze/thaw cycles, which may lead to longer component life.

(18) In some examples, the surface structures (e.g., in the form of protrusions or recessed channels), for example, may be positioned on a surface of an article at the downstream edge of a gas path, upstream of rub path on a seal segment, or distributed along a gas flow surface, or the

like. In some examples, the surface structures may consist of rising and/or narrowing channels that function to draw molten materials (e.g., molten CMAS) up the channels by surface tension and capillary action which will increase gas flow drag on the molten material, pushing it further up to a channel convergence with a sharp back lip that promotes the liberation of molten droplets from the part surface. In some examples, molten materials may be atomized and expelled from a high-temperature mechanical system by the flow of gases across the substrate surface. In some examples, the surface structures are a plurality of channels and/or protrusions, the plurality of channels/protrusions may run parallel to the gas flow but not necessarily perpendicular to a component edge.

(19) Different configurations may allow or benefit from machining the structures into a part surface. In some examples, the structures may be channels which are recessed into a surface coating. The recessed channels may be cooler and less susceptible to erosion, which may reduce thermal stresses in the substrate. Additionally, recessed channels may function collect molten materials, drawing liquid along the channel length by surface tension and capillary action.

Protruding surface features and/or channels rising above the surface plane of a coating may be at higher temperatures which may decrease CMAS viscosity to aid its flow/removal from an article.

(20) The geometry of the surface structures may depend on a number of factors. Surface chemistry of the part, manufacturability, and performance detriment of flow disrupting surface features may be some considerations in selecting the geometry of the plurality of surface structures. In some examples, a plurality of surface structures may be machined into a coated component, additively manufactured, mask and coated, or the like.

(21) In some examples, an article has a surface which is made up of a composite coating on a substrate, where the composite coating is predominantly resistant to molten materials (e.g., CMAS), but incorporates regions of higher susceptibility to molten materials. The susceptible regions may over time be infiltrated and/or react with molten CMAS and removed via spallation and/or dissolved/melted into the CMAS, generating a plurality of surface features (e.g., plurality of channels). In other words, an aspect of the disclosure contemplates a coating having a portion that is designed to be highly susceptible to CMAS attack and another portion that defines plurality of surface structures configured to promote the removal of molten materials along the surface of the substrate when the susceptible portions are removed. Regions of the surface with a coating susceptible to attack (e.g., infiltration of molten materials, reaction, or the like) may react with molten materials, causing spallation of the susceptible coating regions, forming a plurality of surface structures (e.g., a plurality of channels) on the surface. This approach may be particularly useful with an inset plurality of surface features configuration. The susceptible coating may be added in a second coating step to fill in machined or masked channels or joined or manufactured in another way. A coating susceptible to CMAS attack may be more porous than a non-susceptible or CMAS resistant coating region. In some examples, polyester may be included in a thermal spray formulation to create coating regions susceptible to CMAS attack. Alone or in combination, one or more of the following materials may be useful to create a coating that is susceptible to attack by molten materials: MgO, MgAlO, HfSiO, Mo, MoSi.sub.2, SiAlON, Al.sub.2O.sub.3. Conversely, alone or in combination, one or more of the following materials may be useful to create a coating that is non-susceptible to attack by molten materials: GdZrO, YbDS, YbMS, other rare earth silicates, or other refractory materials.

(22) In some examples, the plurality of surface features may function to decrease exposure of the surface to molten materials in the hottest portion of mechanical system, aiding the flow of molten materials downstream (and possibly atomizing molten materials which could be beneficial) to lower temperature areas of a mechanical system where the molten materials CMAS solidifies or can be more easily managed as lower temperatures slow reaction rates.

(23) FIG. 1 is a conceptual schematic diagram illustrating the top view of example article 10 defining outer surface 12 (also referred to as “surface 12”). FIG. 2A is a conceptual schematic

diagram illustrating an example view of article **10** along cross-section A-A. FIG. 2B is a conceptual schematic diagram illustrating an alternative example view of article **10** along cross-section A-A. FIG. 2C is a conceptual schematic diagram illustrating an alternative example view of article **10** along cross-section A-A.

(24) As will be described below, surface **12** of includes a plurality of surface features **14A-14F** (collectively surface features **14**) which define plurality of channels such as channels **24A-24F** (collectively channels **24**). Channels **24** may modify a flow of a liquid such as molten CMAS over the outer surface **12** of the coating **28**, where a gas flows in direction **22** over outer surface **12** of article **10**, e.g., to reduce accumulation of the liquid on outer surface **12** of article **10**. For ease of description, examples of the present disclosure are described with regard to a liquid on outer surface **12** in the form of liquid CMAS although examples of the disclosure may be employed to modify the flow of other liquids over the surface **12** of article **10**.

(25) Article **10** may be a high-temperature component, for example, an industrial, automotive, or aeronautical component. In some examples, article **10** may be a component of a high-temperature mechanical system, such as, for example, a gas turbine engine or the like. In some examples, article **10** may form a portion of a flow path structure, a seal segment, a blade track, blade shroud, an airfoil, a blade, a vane, a combustion chamber liner, or another portion of a gas turbine engine article **10**. While the description herein may be directed to a gas turbine component, it will be understood that the disclosure is not limited to such examples. Rather, plurality of surface features **14** may be included on a surface of any article which requires or may benefit from management of a molten materials flowing over an outer surface of the component. For example, plurality of surface features **14** may be included on a surface of an internal combustion engine, an industrial pump, a housing or internal seal ring of an air compressor, or an electric power turbine. In some examples, as shown, article **10** may have a simple geometric shape. In other examples, article **10** may have a more complicated geometry.

(26) Article **10** includes substrate **16** and coating **28** on substrate **16**. Substrate **12** may include a material suitable for use in a high-temperature environment. In some examples, substrate **12** may include a ceramic or a ceramic matrix composite (CMC) substrate. Suitable ceramic materials, may include, for example, a silicon-containing ceramic, such as silica ( $\text{SiO}_2$ ) and/or silicon carbide (SiC); silicon nitride ( $\text{Si}_3\text{N}_4$ ); alumina ( $\text{Al}_2\text{O}_3$ ); an aluminosilicate; a transition metal carbide (e.g., WC,  $\text{Mo}_2\text{C}$ , TiC); a silicide (e.g.,  $\text{MoSi}_2$ ,  $\text{NbSi}_2$ ,  $\text{TiSi}_2$ ); combinations thereof; or the like. In some examples in which substrate **12** includes a ceramic, the ceramic may be substantially homogeneous. In some examples, substrate **12** may be a reaction bonded silicon carbide (RBSiC) substrate.

(27) In examples in which substrate **12** includes a CMC, substrate **22** may include a matrix material and a reinforcement material. The matrix material may include, for example, silicon metal or a ceramic material, such as silicon carbide (SiC), silicon nitride ( $\text{Si}_3\text{N}_4$ ), an aluminosilicate, silica ( $\text{SiO}_2$ ), a transition metal carbide or silicide (e.g., WC,  $\text{Mo}_2\text{C}$ , TiC,  $\text{MoSi}_2$ ,  $\text{NbSi}_2$ ,  $\text{TiSi}_2$ ), or another ceramic material. The CMC may further include a continuous or discontinuous reinforcement material. For example, the reinforcement material may include discontinuous whiskers, platelets, fibers, or particulates. Additionally, or alternatively, the reinforcement material may include a continuous monofilament or multifilament two-dimensional or three-dimensional weave, braid, fabric, or the like. In some examples, the reinforcement material may include carbon (C), silicon carbide (SiC), silicon nitride ( $\text{Si}_3\text{N}_4$ ), an aluminosilicate, silica ( $\text{SiO}_2$ ), a transition metal carbide or silicide (e.g., WC,  $\text{Mo}_2\text{C}$ , TiC,  $\text{MoSi}_2$ ,  $\text{NbSi}_2$ ,  $\text{TiSi}_2$ ), or the like.

(28) Substrate **12** may be manufactured using one or more techniques including, for example, chemical vapor deposition (CVD), chemical vapor infiltration (CVI), polymer impregnation and pyrolysis (PIP), slurry infiltration, melt infiltration (MI), combinations thereof, or other techniques.

(29) Coating **28** may be an environmental barrier coating (EBC). When coating is in the form of an

EBC, coating **28** may include at least one of a rare-earth oxide, a rare-earth silicate, an aluminosilicate, or an alkaline earth aluminosilicate. For example, coating **28** may include mullite, barium strontium aluminosilicate (BSAS), barium aluminosilicate (BAS), or strontium aluminosilicate (SAS). In some examples, coating **28** may include at least one rare-earth oxide, at least one rare-earth monosilicate ( $\text{RE.sub.2SiO.sub.5}$ , where RE is a rare-earth element), at least one rare-earth disilicate ( $\text{RE.sub.2Si.sub.2O.sub.7}$ , where RE is a rare-earth element), or combinations thereof. The rare-earth element in the at least one rare-earth oxide, the at least one rare-earth monosilicate, or the at least one rare-earth disilicate may include at least one of lutetium (Lu), ytterbium (Yb), thulium (Tm), erbium (Er), holmium (Ho), dysprosium (Dy), gadolinium (Gd), terbium (Tb), europium (Eu), samarium (Sm), promethium (Pm), neodymium (Nd), praseodymium (Pr), cerium (Ce), lanthanum (La), yttrium (Y), or scandium (Sc). Coating **28** may also protect substrate **12** and provide for other functions besides that of environmental protection, e.g., by functioning as a thermal barrier coating (TBC), abradable coating, erosion resistant coating, and/or the like.

(30) When coating **28** functions as a TBC, coating **28** may provide thermal cycling resistance, a low thermal conductivity, erosion resistance, combinations thereof, or the like. In some examples, coating **28** may include zirconia or hafnia stabilized with one or more metal oxides, such as one or more rare earth oxides, alumina, silica, titania, alkali metal oxides, alkali earth metal oxides, or the like. For example, coating **28** may include yttria-stabilized zirconia ( $\text{ZrO.sub.2}$ ) or yttria-stabilized hafnia, or zirconia or hafnia mixed with an oxide of one or more of lutetium (Lu), ytterbium (Yb), thulium (Tm), erbium (Er), holmium (Ho), dysprosium (Dy), gadolinium (Gd), terbium (Tb), europium (Eu), samarium (Sm), promethium (Pm), neodymium (Nd), praseodymium (Pr), cerium (Ce), lanthanum (La), yttrium (Y), or scandium (Sc).

(31) As one example, coating **28** may include a rare earth oxide-stabilized zirconia or hafnia layer including a base oxide of zirconia or hafnia and at least one rare-earth oxide, such as, for example, at least one oxide of lutetium (Lu), ytterbium (Yb), thulium (Tm), erbium (Er), holmium (Ho), dysprosium (Dy), gadolinium (Gd), terbium (Tb), europium (Eu), samarium (Sm), promethium (Pm), neodymium (Nd), praseodymium (Pr), cerium (Ce), lanthanum (La), yttrium (Y), or scandium (Sc). In some such examples, coating **28** may include predominately (e.g., the main component or a majority) the base oxide including zirconia or hafnia mixed with a minority amount of the at least one rare-earth oxide.

(32) Although not shown in FIGS. 2A-2C, article **10** may include one or more optional coatings/layers present between coating **28** and substrate **16**, e.g., in the form of a bond layer (e.g., a silicon-based bond coat) and/or intermediate layer.

(33) Surface **12** of coating **28** may be a gas-washed surface of article **10** with gas flowing in direction **22** during operation of article **10** as a component in a gas turbine engine. In some examples, such as when article **10** is a part of a high temperature mechanical system such as a gas turbine engine, gases flowing over the surface **12** may be hot gases which can melt materials such as CMAS, rendering molten CMAS on surface **12**. As described herein, if allowed to remain in contact with surface **12** (e.g., by pooling or otherwise accumulating on surface **12**), molten CMAS may infiltrate and react with substrate **16**, causing mechanical and/or chemical damage.

(34) Article **10** may be employed with a gas turbine system to define leading edge **18** and trailing edge **20** relative to gases flowing in the direction of arrow **22** through the system. Leading edge **18** may be the point of article **10** that is furthest upstream along the gas-washed flow path, and trailing edge **20** may be the point of article **10** that is furthest downstream along the gas-washed flow path. A gas-washed flow path may thus be defined as the portion of surface **12** between leading edge **18** and trailing edge **20**. In some examples, leading edge **18** and trailing edge **20** may be a component edge.

(35) In accordance with some examples of the disclosure, article **10** includes plurality of surface features **14A-14F**. In the example of FIGS. 1-2C, surface features **14** of article **10** are in the form of

respective protrusions that protrude from surface plane **30** of coating **28**. Surface features **14** are positioned adjacent to trailing edge **20** with the protrusion beginning at position **26** of article **10** and extending downstream to trailing edge **20**. Protruding surface features **14** protrude to height **21** above surface plane **30** at trailing edge **20**. As shown in FIG. 2A, in some examples, protruding surface features **14** may gradually ramp up and away from surface plane **30** while extending in the downstream direction from position **26** to trailing edge **20**. Additionally, or alternatively, one or more of protruding surface features **14** may protrude relatively abruptly at position **26** (e.g., in a stepwise fashion), as shown in FIG. 2B, and may gradually ramp away from surface plane **30** moving in the downstream direction towards trailing edge **20**.

(36) Plurality of channels **24A-24F** are defined between protruding surface features **14**. Channels **14** may be recessed relative to protruding surface features **14** (e.g., as shown in FIGS. 2A-2C for channel **24E** formed between the vertically extending walls of protruding surface features **14E** and **14F**). In some examples, the bottom surface of channels **14** may be substantially even with surface plane **30**. In some examples, such as that shown in FIGS. 2A-2C, channels **14** may ramp up from surface plane **30** moving downstream, e.g., to height **23** in FIG. 2A.

(37) The angular nature of protruding surface features **14** results in the width of channels **24** being tapered, e.g., in direction **22**. In the examples of FIGS. 1-2C, channels **14** taper such that the width along the x-direction indicated in FIG. 1 narrows moving in the downstream direction. In such a configuration, the inlet openings of respective channels **14** at position **26** have a greater width than the outlet openings adjacent to trailing edge **20** (e.g., at outlet opening **31** for channel **24E**).

(38) Protruding surface features **14** and channels **24** may function to modify the flow of molten materials over surface **12** of coating **28** in direction **22**, e.g., in the presence of a gas flow during operation of article **10** as a component in a gas turbine engine that operates in a high temperature environment. For example, the rising and/or narrowing nature of channels **14** may function to draw molten CMAS up channels **14** by surface tension and/or capillary action which will increase gas flow drag on the fluid/liquid CMAS pushing the CMAS further up to a channel convergence (e.g., outlet opening **31**) with a sharp back lip at trailing edge **20** that promotes the liberation of molten droplets from surface **12** of article **10**. In some examples, the incline/ramp of channels **24** and/or protruding features **14** may assist in wicking molten CMAS to better transporting molten CMAS to cooler zone of the engine downstream of article **10** and/or atomize and exhaust material by increasing fluid drag on molten CMAS in a channel **24**. For instance, molten material at the top of the ramp of channel **24E** at outlet opening **31** at trailing edge **20** may be exposed to high velocity gases flowing along the gas path, which may atomize molten material. Moreover, channels protruding (e.g., in a ramping fashion) from surface plane **30** may be at higher temperatures during operation of a gas turbine engine, which may decrease molten CMAS viscosity to aid its flow/removal from article **10** downstream of trailing edge **20**. Accordingly, surface features **14** and channels **24** on the gas washed surface of article **10** may promote the removal of molten CMAS from surface **12**.

(39) Channels **24** and protruding surface features **14** may have any suitable shapes and dimensions. In some examples, it may be desirable for protruding surface features **14** to have height **21** that is relatively small to prevent huge protrusion into the gas stream over surface **12** but still enough to function as described herein. In some examples, height **21** may be approximately equal to the max depth at which CMAS may accumulate or pool on surface **12** absent protrusions **14** and channels **24**. In some examples, height **21** may be up to about 2 millimeters (mm) such as greater than zero to 2 mm. Height **23** may be equal to or greater than about 20 microns, such as at least about 100 microns, at least about 120 microns, or up to 2 millimeters, such as height **23** being about zero, greater than zero and/or less than or equal to 2 mm. Height **23** of channels **24** may be such that the difference between height **21** and height **23** is approximately 100 microns or less, such as from 100 microns to zero (e.g., where height **21** is equal to height **23**). The shape/dimensions of individual channels **24** may vary between channels or may be substantially the same for all channels. The



shape/dimensions of individual protruding surface features **14** may vary between respective protruding surface features or may be substantially the same for all protruding surface features **14**. (40) FIG. 2C illustrates an example cross-section of article **10** along A-A where height **21** of protrusion **14E** is approximately equal to height **23** of channel **24E**. In some an example, the ramp of channel **24** from the surface plane **30** starts downstream of location **26** where the ramp up of protruding surface feature **14E** begins. In such an example, the angle of the ramp up for channel **24E** is greater than the ramp angle of channel **14E**, where the apex of channel **24E** and the apex of protruding surface feature **14E** are equal to each other, e.g., directly adjacent to trailing edge **20**.

(41) The inlet width (in the x-direction of FIG. 1) of respective channels **24** at the most upstream position as defined by the starting of protruding surface features **14**, e.g., at location **26** in FIG. 1, may be any suitable value. In some examples, the inlet width may be fairly wide and dependent on the overall dimensions of article **10**. In some examples, the inlet width may be at least about 0.5 millimeters, such as about 0.5 mm to about 5 mm. The outlet width of the respective channels **24** (e.g., the width of channels **24** further downstream) may be less than the inlet width. In some examples, the outlet width (e.g., directly adjacent to trailing edge **20**) may be about 100 microns to about 1 mm. In some examples, channels **24** converge to a zero width at the outlet, e.g., where height **23** and height **21** are approximately equal.

(42) The length of channels **24** and protruding surface features **14** (in the gas flow direction **22**) may be dependent on the overall length of article **10**. Channels **24** and protruding surface features **14** may be located on article **10** to coincide/overlap with an area on surface **12** that would otherwise be susceptible to CMAS attack (e.g., from CMAS pooling). For example, location **26** on article **10** may be just upstream of a location on surface **12** of article **10** that would otherwise be susceptible to CMAS attack. Channels **24** and protruding surface features **14** may extend over the entire area that is susceptible to CMAS attack, e.g., to help prevent accumulation of CMAS in that area by assisting the flow of molten CMAS downstream from location **26** beyond the furthest downstream portions of channels **24** and protruding surface features **14**. In one example where the length of surface **12** that would be susceptible to CMAS attack may be about 1 cm, protruding surface features **14** and/or channels **24** may be greater than 1 cm, such as 2 cm, and may overlap with the 1 cm portion on surface **12**, e.g., where the “mouth” of channels **24** are at the area susceptible to CMAS attack.

(43) While channels **24** and protruding surface features **14** are shown extending from a position down stream of leading edge **18** of article **10**, starting at position **26** upstream of trailing edge **20**, channels **24** and protruding surface features **14** may be positioned at any suitable location on surface **12**, e.g., in the area of article **10** where coating **28** is most susceptible to CMAS attack. In some examples, channels **24** and protruding surface features **14** end prior to trailing edge **20**. In some examples, rather than a single “row” of channels **24** and protruding surface features **14**, article **10** may include multiple “rows” arranged one after another in the gas flow direction **22**, e.g., where the size of article **10** in flow direction **22** allows for multiple rows.

(44) The examples of FIGS. 1 to 2C include surface features that protrude from the surface plane of a coating to define channels that at or above the surface plane of the coating. In other examples, a coating may include surface features in the form of a plurality of channels which are recessed into surface of a coating.

(45) FIGS. 3 and 4 are conceptual diagrams illustrating such an example. FIG. 3 is a conceptual diagram illustrating the top view of an example article **40** including coating **28** on substrate **16**. FIG. 4 is the conceptual diagram illustrating the example article **40** of FIG. 3 along cross section B-B. Article **40** and article **10** may be similar to each other and like features are similarly numbered. For example, coating **28** and substrate **16** of article **40** may be substantially the same as coating **28** and substrate **16** of article **10** in FIGS. 1-2C.

(46) However, unlike example article **10**, coating **28** includes a plurality of channels **54A-54F** (collectively “channels **54**”) recessed into outer surface **12** of coating **28**. In the example of FIGS. 3

and **4**, channels **54A-54F** are defined in part by surface features **44A-44F**, where the recessed nature of channels **54** results in surface feature **44A-44F** extending away (e.g., in a finger-like fashion) from position **26** on surface **12** of coating **28** with a height that is approximately along the same plane as the plane of surface **12**. In other examples, surface features **44** may protrude from the plane of surface **12** of coating **12** (e.g., similar to that of features **14** of article **10**) in combination with channels **54** being recessed into surface **12**.

(47) In the illustrated example of article **40**, channels **54** have a ramped configuration (e.g., as illustrated in FIG. **4** for channel **54E**) in direction **22** where the depth of recess into coating increases moving from position **26** to trailing edge **20**. The vertical walls of recessed channels **54** define the perimeters of surface features **44A-44F**. In the example of FIGS. **3** and **4**, the ramp down for recessed channels **54** may start at the plane of surface **12** at position **26** and gradually ramp down in a linear manner. In other examples, the ramp may be in a stepwise fashion, e.g., with a single stepwise drop into surface **12** at position **12**, and a bottom surface of channels **54** that extends substantially co-planar with the plane of surface **12** or with a ramp to further increase the depth of recess into plane of surface **12**.

(48) The width along the x-axis shown in FIG. **3** of channels **54** may taper moving in the downstream direction. In the example article **40** of FIGS. **3** and **4**, the taper is such that the width increases moving from position **26** to trailing edge **20**. Channels **54** each define an outlet opening at trailing edge **20**, such as outlet opening **46** for channel **54E**.

(49) Channels **54** may have any suitable dimensions and shape, e.g., that allows channels **54** to function as described herein. In some examples, channels **54** may be the inverse dimensions of channels **24** described above (e.g., with the height **23** for channels **24** being the max depth of channels **54**, and the inlet width and outlet widths of channels **24** being the widest and narrowest width, respectively, of channels **54**). In some examples, channels **54** may have a depth into surface **12** that is greater than zero but less than the thickness  $t$  of coating **28** as labelled in FIG. **4**. In some examples, thickness  $t$  of coating **28** may be about 100 microns to about 2 mm. In some example, channels **54** are recessed to a maximum depth of about 50% of the thickness  $t$  of coating **28**.

(50) As described herein, recessed channels **54** may be inset into surface **12** of coating **28**. The recessed nature of channels **54** may be desirable because recessed portion may be relatively cooler and less susceptible to erosion. Additionally, thermal stresses may be reduced by recessed channels **54**. Although the volume defined by recessed channels **54** may be below the plane of surface **12** upstream of position **26** and less influenced by the flow of gas along direction **22**, the narrowed and tapered nature of channels **54** at position **26** may function to provide a capillary pull on liquid CMAS entering channels **54** towards trailing edge **20**. Moreover, the length **47** of channels **54** may be great enough to still allow for the flow of gas along direction **22** over surface **12** to force a liquid in channels **54** along direction **22** and out of channels at trailing edge **20** (e.g., out of outlet opening **46**). As such, recessed channels **54** may modify the flow of molten CMAS over surface **12**, e.g., to promote the removal of the CMAS from article **40** in a downstream direction beyond trailing edge **20**.

(51) As describe herein, channels **24** and channels **54** may be defined by a pair of corresponding vertical walls either protruding from surface **12** (as in the case of channels **24**) or recessed into surface **12** (as in the case of channels **54**). The vertical walls of the channels as well as the bottom surface of the channels may define flow path for molten materials (CMAS) to flow along in direction **22**, e.g., starting at position **26** on surface **12**. As described herein, in some examples, the width (along x-axis labelled in FIG. **1**) of channels **24** and channels **54** may be tapered in flow direction **22**. For example, as shown in FIG. **1**, channel **24E** narrows from position **26** to a convergence at outlet opening **31**, with channel **24E** being widest at position **26** and narrowest downstream adjacent to trailing edge **31**. Conversely, channel **54E** diverges/widens from position **26** to outlet opening **46**, with channel **54E** being narrowest at position **26** and widest downstream adjacent to trailing edge **20**. In still other examples, channels **24** and **54** may have a uniform or

nearly uniform width throughout. In some examples, each channel in the plurality of channels may have a geometry substantially similar to channels **24** or channels **54** although other geometries are contemplated. In some examples, a plurality of channels may combine with other surface features to form plurality of surface features **14** (e.g., other ramps, recesses, slots, or the like configured to manage molten materials on surface **12** of substrate **16**).

(52) In some examples, as shown, channels **24**, **54** and surface features **14**, **44** are positioned to extend from trailing edge **20** of articles **10**, **40** at least partially upstream to position **26** nearer leading edge **18**. Additionally, or alternatively, in some examples, plurality of surface features **14**, **44** and/or channels **24**, **54** may extend from adjacent to leading edge **18** at least partially across surface **12** toward trailing edge **20**. In some examples, surface features **14**, **44** and/or channels **24**, **54** may extend across substantially all of surface **12** from leading edge **18** to trailing edge **20**, or may be positioned at discrete locations along surface **12** of coating **28**. Channels **24**, **54** and surface features **14**, **44** may be orientated with regard to direction **22** of gas flow during operation of a gas turbine engine system that includes articles **10**, **40** as a component (e.g., as a turbine blade, vane, or shroud).

(53) FIG. **5** is a conceptual diagram illustrating the top view of another example article **50** according to the present disclosure. FIG. **6** is a conceptual diagram illustrating example article **50** of FIG. **5** along cross-section C-C. Article **50** may be similar to articles **10** and **40** and like features are similarly numbered. For example, coating **28** and substrate **16** of article **50** may be substantially the same as coating **28** and substrate **16** of articles **10** and **40** in FIGS. **1-4**.

(54) In the example of FIGS. **5** and **6**, coating **28** includes multiple layers; EBC layer **28B** on substrate **16** and abradable layer **28A** on EBC layer **28B**. In other examples, layer **28B** may be a TBC layer. Abradable layer **28A** may have the same or substantially similar composition to that described herein for an EBC but with a greater porosity than, e.g., EBC layer **28B**. However, abradable layer **28A** may be configured to be abraded by an opposing component during operation of a gas turbine engine with article **50** being incorporated as a component (e.g., as a blade track). For example, with article **50** forming a blade track, abradable layer **28A** may be abraded by interaction with the tip of an opposing, rotating blade. The opposing blade tip may abrade into surface plane **30** along a path to cut track **60** into coating **28** to a depth **D**. In the example of FIGS. **5** and **6**, the path of the opposing blade tip that forms track **60** may be in x-direction indicated in FIG. **5** (e.g., perpendicular to the flow direction **22**). As illustrated by the dashed line, surface **12** of coating **28A** may be a flat or substantially level surface as initially manufactured, but configured to be abraded during use, such as part of a seal segment of a gas turbine engine.

(55) As shown in FIGS. **5** and **6**, article **50** includes protruding surface features **14** positioned upstream of track **60**, e.g., extending from directly adjacent of upstream edge **18** to directly adjacent to track **60**. Although not labelled in FIGS. **5** and **6**, like that in FIGS. **1-2C**, neighboring protruding surface features **14** may define respective channels (such as channels **24**) on surface **12** of coating **28**. Protruding surface features **14** and its associated channels are oriented to the direction of gas flow **22**.

(56) The protruding surface features **14** and/or the channels may function to modify the flow of molten CMAS over surface **12**, e.g., to promote the movement of the liquid CMAS over surface to downstream of trailing edge **20**. For example, the channels defined by protruding surface features **14** are widest at the upstream end and narrows to a convergence at the opposing downstream end. Molten materials may collect and be fed into wider opening of the channels, with the upstream end of the channels acting as a channel mouth. Molten materials in the channel may wick or otherwise flow towards the downstream end through surface tension and/or capillary action, pooling deeper as the channels narrows to a convergence at the end adjacent to track **60**. Gas flow drag may increase as molten materials climb the incline of the channels because gas velocity may increase further away from surface **12**. Thus, fluid drag from gases flowing over the surface of the channels may increase as molten materials travel along the channels, promoting liberation of molten

materials from surface **12** of coating **28A**. The protrusions and channels in FIGS. **5** and **6** may have the same shape and/or dimensions as described herein for channels **24** and protrusions **14** in FIGS. **1-2C**.

(57) As shown, article **50** also includes recessed channels **64A-64F** (collectively “channels **64**”). Channels **64** are located adjacent to the downstream edge of track **60** and are recessed into plane **30** of coating **28A** to a max depth **D** approximately equal to the maximum cut depth of track **60**. The depth of channels **64** ramp up from depth **D** moving towards trailing edge **64**. In the example illustrated in FIG. **6**, the ramp up in depth of channels **64** is such that the end of channels **64** at trailing edge **20** is approximately equal to (e.g., flush or nearly flush with) surface plane **30** of coating **28A**. In other examples, channels **64** may ramp up to a height above surface plane **30** or to a depth still below surface plane **30** at a position nearest trailing edge **20**. Channels **64** are oriented to the direction **22** of gas flow over surface **12**. Channels **64** may function to provide a pathway for molten CMAS to be removed from the bottom of track **60** (e.g., from the bottom of a rubbed abradable seal segment).

(58) Unlike that of recessed channels **54** in FIGS. **3** and **4**, channels **64** are tapered to widen in width moving downstream along direction **22**. This may allow molten CMAS within track **60** to collect within the wide mouth of channels **64** and then be drawn up the ramp and concentrated within channels **64** as channels **64** narrow in width moving downstream. In other examples, some or all of channels **64** are configured to taper in the opposite direction with the widest portion being downstream, e.g., adjacent to trailing edge **20**. In other examples, some or all of channels **64** do not exhibit a tapering width in the downstream direction **22**.

(59) While the channels **64** are shown extending from track **60** to trailing edge **20**, in other examples, channels **64** may not extend all the way to trailing edge **20**. Likewise, while protruding surface features **14** are shown extending from leading edge **18** to track **60**, in other examples, channels **64** may not extend all the way to leading edge **18** and/or to track **60**.

(60) In the examples of FIGS. **1-6**, leading edge **18** and trailing edge **20** are both substantially perpendicular to gas flow direction **20**. In such examples, the surface features and channels may extend in a direction substantially parallel to the gas flow direction and substantially perpendicular to leading edge **18** and/or trailing edge **20**. FIG. **7** is conceptual diagram illustrating an example article **70** in which the leading edge **18** and trailing edge **20** are not perpendicular to direction **22** of the gas flow over surface **12** of coating **28**. As shown in FIG. **7**, in such as example, protruding surface features **14** and channels **24** may be orientated such that the central axis (indicated as a dashed line for each) extends in a direction substantially parallel to direction **22** of the gas flow and not perpendicular to trailing edge **20** or lead edge **18**. Thus, in examples in which the gas flow path does not align with a trailing/leading edge of a component, it may be desirable to orient surface features **14** and/or channels **24** to be substantially parallel to the direction **22** of gas flow rather than with an edge of a component (e.g., a CMC gas turbine component) if the gas flow will not be perpendicular to the part edge, to promote removal of molten materials from surface **12** of a coating **28** in the manner described herein.

(61) While the examples of FIGS. **1-7** are shown with article **10** have a rectangular shaped surface **12**, examples are not limited to such surface shapes. In some examples, surface **12** of article **10** may increase or decrease in width, e.g., moving in the gas flow direction **22**. In some examples, article **10** may have a trapezoidal surface shape with an increasing width in the gas flow direction **22**, e.g., in the case of a turbine vane. Additionally, surface **12** is not limited to a flat surface plane. For examples, the surface of article **10** may have a curvilinear or non-planar profile, e.g., in the direction of gas flow **22**.

(62) Any suitable techniques may be employed to form articles in accordance with the disclosure, such as articles **10**, **40**, **50**, and **70** including the described surface features/channels. FIG. **8** is a flow diagram illustrating one example technique for manufacturing such an article, e.g., for use as a component in a gas turbine engine. The technique of FIG. **8** is described with reference to article **30**

of FIGS. 3 and 4. However, a person having skill in the art will understand that the technique of FIG. 8 may be performed to manufacture another example article, such as articles 10, 50, and 70.

(63) As shown, coating 28 may be formed on substrate 16 using any suitable technique (80). In some examples, coating 28 may be formed by a thermal spray process, a slurry deposition process, and/or additive manufacturing process. In the example of FIG. 8, as initially deposited, coating 28 may have a substantially planar surface 12.

(64) Subsequently, channels 54 may be formed in coating 28 (82). Any suitable process may be employed such as a material removal process. In some examples, a machining process (or other suitable selective removal process) may be employed to selectively remove portions of coating 28 to form recessed channels 54. The removal of portions of coating 28 to form recessed channels 54 may also result in the formation of surface features 44 extending from position 26.

(65) In other examples, in addition to, or as an alternative to formation of a coating and selective removal of the coating, other techniques may be employed including suitable additive manufacturing techniques and techniques that iteratively deposit portions of material with selective masking (e.g., to allow for selective build up and/or removal of material).

(66) In one example, the example surface features and channels of articles 10, 40, 50, and 79 may be achieved by initially forming a coating on substrate 16 with portions of the coating more susceptible to failure and removal from coating 28 during the operating life of the component. For example, in the case of recessed channels 54 inset in coating 28 for article 40, rather than being open, channels 54 of article 40 may initially be filled with a coating material that is different from the remaining portions of coating 28 shown in FIGS. 3 and 4. The material that initially “fills” channels 54 may be more susceptible to CMAS attack compared to the remaining portions of coating 28. In such an example, during the operating life of article 40, the CMAS susceptible portions may be infiltrated or otherwise caused to fracture and be removed from coating 28. Upon remove, open channels 54 may be formed coating 28, which may function as described herein to promote removal of molten CMAS from article 40. In other example, in addition to, or as an alternative to CMAS attack, the susceptible portions of coating 40 may be more susceptible to erosion or other mechanism that causes those portions to be removed during the operating life of article 40 within a gas turbine engine.

(67) As such, in some examples, coating 28 on article 40 may initially include first and second portions, where the first portion is more susceptible to removal (e.g., via CMAS attack) compared to the second portion. In some examples, the first portion may be more susceptible to CMAS attack based on the composition and/or microstructure of the first portion compared to the second coating. For example, the composition of first portion may include one or more of MgO, MgAlO, HfSiO, Mo, MoSi.sub.2, SiAlON, or Al.sub.2O.sub.3. Conversely, the composition of second portion may include GdZrO; a RE silicate such as yttrium monosilicate or other RE monosilicate and/or yttrium disilicate; or other refractory metals. In some examples, the composition of first portion may be formed almost entirely of one or more of MgO, MgAlO, HfSiO, Mo, MoSi.sub.2, SiAlON, or Al.sub.2O.sub.3, e.g., with the second portion containing less or none of the material that forms the first portion. In some examples, first and second portion are formed of GdZrO; a RE silicate such as yttrium monosilicate or other RE monosilicate and/or yttrium disilicate; or other refractory metals, but with the first portion including an additive such as MgO, MgAlO, HfSiO, Mo, MoSi.sub.2, SiAlON, and/or Al.sub.2O.sub.3, or a higher concentration of such an additive compared to the second portion. In some examples, the first portion may include at least about 5 weight percent (wt %), such as at least about 10 wt %, or up to 40 wt % of such an additive.

(68) Additionally, or alternatively, the material of the first portion may have a higher void volume (e.g., being more porous) than compared to the material of the second portion. Such microstructure may be tailored by employing sacrificial elements within the composition of the first portion (e.g., in the form of polyester) that may be burned out during a subsequent processing step to form a relatively porous structure for the first portion and not the second portion. In some examples, one

or more parameters of a deposition process such as a thermal spray process may be tailored to provide for the desired microstructure. Thus, the first portion may be made to be more susceptible to removal (e.g., based on CMAS attack) based on the chemistry/composition and/or the microstructure of the first portion compared to the second portion.

## EXAMPLES

(69) The interaction of molten CMAS on the outer surface of a coating of a component within a gas turbine engine was investigated. FIGS. **9-12** are images illustrating component with coatings **128** with various resistance to CMAS and levels of exposure to CMAS. As noted herein, a molten CMAS material **109** may collect on one or more outer surfaces of a component of a high temperature system. FIG. **9** is an image showing a CMAS resistant coating **128** for a CMC component of a gas turbine engine which has been exposed to molten CMAS **109** for **8** hours. The composition of coating **128** in FIGS. **9** and **10** was predominantly ytterbium disilicate with some amount of ytterbium monosilicate.

(70) As shown, the CMAS **109** is pooled on the surface **112** of coating **128**, slowly reacting and not penetrating the coating after **8** hours. The image of FIG. **9** demonstrates that CMAS and other molten materials, in some examples, require extended time on a surface of a coating (such as surface **12** of coating **28** for article **10** of FIG. **1**) to react with the coating and cause deleterious effects. Promoting wicking or other flow of molten materials along the outer surface through the use of surface features/channels, such as the various example surface features and channels described herein, may assist in transporting molten CMAS to cooler downstream regions within the system. In the cooler regions, reaction rates may be relatively slower before the molten CMAS can react with an underlying barrier coating. Deleterious effects of CMAS and/or other molten materials may thus be reduced.

(71) FIG. **10** is an image illustrating the CMAS resistant barrier coating of FIG. **9** after **96** hours, showing the molten CMAS **109** beginning to react with and penetrate the coating **128**. As shown, allowing molten CMAS **109** to contact the coating **128** for a longer period of time allows the CMAS **109** to further penetrate and react with the coating **128**. FIGS. **9** and **10** demonstrate that reaction of molten materials with a substrate or substrate coating may be time dependent, and reducing the time molten materials interact with a surface of a coating such as surface **12** of articles **10**, **40**, and **50**, by promoting transport of molten materials along surface **12** to cooler regions of a high temperature system may be desirable to preserve substrate or coating life.

(72) FIG. **11** is an image illustrating an example CMAS susceptible coating **128** with the same amount of molten CMAS as shown in FIG. **9** but with the CMSA absorbed into the coating **128**. As shown, the coating **128** allows molten CMAS to penetrate into surface **112** the coating **128**. In the example of FIGS. **11** and **12**, coating **128** was a magnesium aluminate (MgAlO) layer.

(73) FIG. **12** is a magnified image illustrating the CMAS susceptible coating of FIG. **11**, showing a first portion **90** displaying an open microstructure (e.g., porosity) on the right side and a second portion **92** in which the coating (e.g., fully) penetrated by molten CMAS within the microstructure on the left side. The created microstructure may be prone to cracking and spallation, and may have reduced structural strength relative to a CMAS resistant coating.

(74) As described above, in some examples, a coating may include one or more CMAS susceptible portions when an article is manufactured. During the operating life of the article, the susceptible portions may fail and be removed from the coating to define a plurality of surface features/channels formed by the failure or other removal of the susceptible portions of the coating. As the CMAS susceptible coating is penetrated and reacted, the microstructure is created, the susceptible coating may crack and fall off, revealing the plurality of surface features/channels below. In this way, the surface features/channels may be on the part are initially but not revealed on the outer surface of a coating when the article is new, but may be exposed after the coating selectively fails, e.g., during the operating life of the article.

(75) Various examples have been described. These and other examples are within the scope of the

following clauses and claims.

(76) Clause 1. An article comprising: a ceramic or ceramic matrix composite (CMC) substrate; and a coating on the ceramic or the CMC substrate, wherein the coating defines an outer surface of the article, wherein the coating includes a plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article.

(77) Clause 2. The article of clause 1, wherein the plurality of surface features includes one or more protrusions protruding out of a surface plane of the coating, and wherein the one or more protrusions define the channels on the outer surface of the article.

(78) Clause 3. The article of clause 2, wherein respective channels on the outer surface of the coating are defined by neighboring individual protrusions of the one or more protrusions.

(79) Clause 4. The article of clause 3, wherein the respective channels have a ramped surface between the neighboring individual protrusions.

(80) Clause 5. The article of clause 1, wherein the plurality of surface features includes one or more recesses recessed into a surface plane of the coating, wherein the one or more recesses define the channels on the outer surface of the article.

(81) Clause 6. The article of any one of clauses 2-5, wherein the channels are tapered in a direction of the gas flow over the outer surface.

(82) Clause 7. The article of any one of clauses 1-6, wherein at least a portion the plurality of surface features are located at a trailing edge of the article relative to the gas flow.

(83) Clause 8. The article of any one of clauses 1-6, wherein the coating includes an abradable coating, and wherein at least a portion the plurality of surface features are located adjacent a track formed into the abradable coating by the abrasion of an opposing abrasive member.

(84) Clause 9. The article of clause 8, wherein a first set of surface features of the plurality of surface features are located adjacent to the track at an upstream edge of the track, and wherein a second set of surface features of the plurality of surface features are located adjacent to the track at a downstream edge of the track.

(85) Clause 10. The article of clause 9, wherein the first set of surface features includes a plurality of protrusions protruding out of a surface plane of the article, and the second set of surface features includes a plurality of recesses recessed into the surface plane of the article.

(86) Clause 11. The article of any one of clauses 1-10, wherein the coating includes a first portion that is more susceptible to CMAS attack as compared to a second portion of the coating, wherein the first portion is configured to, during operation of the article in a high temperature environment, be removed from the coating to define at least a portion of the plurality of surface features with the second portion.

(87) Clause 12. The article of clause 11, wherein the first portion is more susceptible to CMAS attack based on at least one of a microstructure of the first portion or a composition of the first portion as compared to the second portion.

(88) Clause 13. The article of any one of clauses 1-12, wherein the article is a component of a gas turbine engine configured to operate in a high temperature environment in which the molten CMAS is present.

(89) Clause 14. The article of clause 13, wherein the component comprises at least one of a turbine blade, vane, or shroud.

(90) Clause 15. The article of any one of clauses 1-14, wherein the plurality of surface features are configured to modify the flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating by at least one of accumulating the molten CMAS in the channels, raise the molten CMAS above the surface plane of the coating into the gas flow, or drawing the molten CMAS to a trailing edge of the article.

(91) Clause 16. The article of any one of clauses 1-15, wherein the coating includes an environmental barrier coating including a rare earth silicate.

(92) Clause 17. A method comprising forming a coating on a ceramic or ceramic matrix composite (CMC) substrate, wherein the coating defines an outer surface of the article, wherein the coating includes a plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article.

(93) Clause 18. The method of clause 17, wherein the plurality of surface features includes one or more protrusions protruding out of a surface plane of the coating, and wherein the one or more protrusions define the channels on the outer surface of the article.

(94) Clause 19. The method of clause 18, wherein respective channels on the outer surface of the coating are defined by neighboring individual protrusions of the one or more protrusions.

(95) Clause 20. The method of clause 19, wherein the respective channels have a ramped surface between the neighboring individual protrusions.

(96) Clause 21. The method of clause 17, wherein the plurality of surface features includes one or more recesses recessed into a surface plane of the coating, wherein the one or more recesses define the channels on the outer surface of the article.

(97) Clause 22. The method of any one of clauses 18-21, wherein the channels are tapered in a direction of the gas flow over the outer surface.

(98) Clause 23. The method of any one of clauses 17-22, wherein at least a portion the plurality of surface features are located at a trailing edge of the article relative to the gas flow.

(99) Clause 24. The method of any one of clauses 17-22, wherein forming the coating includes forming an abradable coating, and wherein at least a portion the plurality of surface features are located adjacent a track formed into the abradable coating by the abrasion of an opposing abrasive member.

(100) Clause 25. The method of clause 24, wherein a first set of surface features of the plurality of surface features are located adjacent to the track at an upstream edge of the track, and wherein a second set of surface features of the plurality of surface features are located adjacent to the track at a downstream edge of the track.

(101) Clause 26. The method of clause 25, wherein the first set of surface features includes a plurality of protrusions protruding out of a surface plane of the article, and the second set of surface features includes a plurality of recesses recessed into the surface plane of the article.

(102) Clause 27. The method of any one of clauses 17-26, wherein the coating includes a first portion that is more susceptible to CMAS attack as compared to a second portion of the coating, the method comprising removing the first portion from the coating during operation of the article in a high temperature environment to define at least a portion of the plurality of surface features with the second portion.

(103) Clause 28. The method of clause 27, wherein the first portion is more susceptible to CMAS attack based on at least one of a microstructure of the first portion or a composition of the first portion as compared to the second portion.

(104) Clause 29. The method of any one of clauses 17-28, wherein the article is a component of a gas turbine engine configured to operate in a high temperature environment in which the molten CMAS is present.

(105) Clause 30. The method of clause 29, wherein the component comprises at least one of a turbine blade, vane, or shroud.

(106) Clause 31. The method of any one of clauses 17-30, wherein the plurality of surface features are configured to modify the flow of the molten CMAS over the outer surface of the coating by at least one of accumulating the molten CMAS in the channels, raise the molten CMAS above the surface plane of the coating into the gas flow, or drawing the molten CMAS to a trailing edge of the



article.

(107) Clause 32. The method of any one of clauses 17-31, wherein the coating includes an environmental barrier coating including a rare earth silicate.

## Claims

1. An article comprising: a ceramic or ceramic matrix composite (CMC) substrate; and a coating on the ceramic or the CMC substrate, wherein the coating defines an outer surface of the article, wherein the coating includes a first portion which does not include surface features and a second portion adjacent to the first portion, the second portion defining a plurality of surface features, the plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article, wherein: the plurality of surface features includes one or more protrusions protruding out of a surface plane of the coating, wherein the surface plane of the coating is defined by the first portion of the coating, and wherein the one or more protrusions define the channels on the outer surface of the article, respective channels on the outer surface of the coating are defined by neighboring individual protrusions of the one or more protrusions, and the respective channels have a ramped surface between the neighboring individual protrusions, wherein the ramped surface is a planar surface and defines a first point and a second point on the planar surface, wherein the first point and the second point are at different distances away from the surface plane of the coating along a line substantially parallel to the gas flow over the outer surface of the article, wherein the ramped surface defines an elevation change in a Z-direction relative to the surface plane of the coating such that the respective channels are recessed into or protrude out of the surface plane of the coating.
2. The article of claim 1, wherein the plurality of surface features includes one or more recesses recessed into the surface plane of the coating, wherein the one or more recesses define the channels on the outer surface of the article.
3. The article of claim 1, wherein the channels are tapered in a direction of the gas flow over the outer surface.
4. The article of claim 1, wherein at least a portion the plurality of surface features are located at a trailing edge of the article relative to the gas flow.
5. The article of claim 1, wherein the coating includes an abradable coating, and wherein at least a portion the plurality of surface features are located adjacent a track formed into the abradable coating by the abrasion of an opposing abrasive member.
6. The article of claim 5, wherein a first set of surface features of the plurality of surface features are located adjacent to the track at an upstream edge of the track, and wherein a second set of surface features of the plurality of surface features are located adjacent to the track at a downstream edge of the track.
7. The article of claim 6, wherein the first set of surface features includes a plurality of protrusions protruding out of a surface plane of the article, and the second set of surface features includes a plurality of recesses recessed into the surface plane of the article.
8. The article of claim 1, wherein the coating includes a first portion that is more susceptible to CMAS attack as compared to a second portion of the coating, wherein the first portion is configured to, during operation of the article in a high temperature environment, be removed from the coating to define at least a portion of the plurality of surface features with the second portion.
9. The article of claim 8, wherein the first portion is more susceptible to CMAS attack based on at least one of a microstructure of the first portion or a composition of the first portion as compared to the second portion.
10. The article of claim 1, wherein the article is a component of a gas turbine engine configured to

operate in a high temperature environment in which the molten CMAS is present.

11. The article of claim 10, wherein the component comprises at least one of a turbine blade, vane, or shroud.

12. The article of claim 1, wherein the plurality of surface features are configured to modify the flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating by at least one of accumulating the molten CMAS in the channels, raising the molten CMAS above the surface plane of the coating into the gas flow, or drawing the molten CMAS to a trailing edge of the article.

13. The article of claim 1, wherein the coating includes an environmental barrier coating including a rare earth silicate.

14. A method comprising forming a coating on a ceramic or ceramic matrix composite (CMC) substrate on an outer surface of an article, wherein the coating includes a first portion which does not include surface features and a second portion adjacent to the first portion, the second portion defining a plurality of surface features, the plurality of surface features defining channels on the outer surface of the article, wherein the channels are configured to modify a flow of molten Calcia-Magnesia-Alumina Silicate (CMAS) over the outer surface of the coating in a gas flow over the outer surface of the article to reduce accumulation of the molten CMAS on the outer surface of the article, wherein: the plurality of surface features includes one or more protrusions protruding out of a surface plane of the coating, wherein the surface plane of the coating is defined by the first portion of the coating, and wherein the one or more protrusions define the channels on the outer surface of the article, respective channels on the outer surface of the coating are defined by neighboring individual protrusions of the one or more protrusions, and the respective channels have a ramped surface between the neighboring individual protrusions, wherein the ramped surface is a planar surface and defines a first point and a second point on the planar surface, wherein the first point and the second point are at different distances away from the surface plane of the coating along a line substantially parallel to the gas flow over the outer surface of the article, wherein the ramped surface defines an elevation change in a Z-direction relative to the surface plane of the coating such that the respective channels are recessed into or protrude out of the surface plane of the coating.

15. The method of claim 14, wherein the plurality of surface features includes one or more recesses recessed into the surface plane of the coating, wherein the one or more recesses define the channels on the outer surface of the article.

16. The method of claim 14, wherein the channels are tapered in a direction of the gas flow over the outer surface.

17. The method of claim 14, wherein at least a portion the plurality of surface features are located at a trailing edge of the article relative to the gas flow.

18. The method of claim 14, wherein the coating includes an abradable coating, and wherein at least a portion the plurality of surface features are located adjacent to a track formed into the abradable coating by the abrasion of an opposing abrasive member.

19. The method of claim 18, wherein a first set of surface features of the plurality of surface features are located adjacent to the track at an upstream edge of the track, and wherein a second set of surface features of the plurality of surface features are located adjacent to the track at a downstream edge of the track.

20. The method of claim 19, wherein the first set of surface features includes a plurality of protrusions protruding out of a surface plane of the article, and the second set of surface features includes a plurality of recesses recessed into the surface plane of the article.

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